

TinyNPU: A Sensor-Aware Reconfigurable AI Accelerator for Ultra-High Speed Ballistic Vision

19EC702 Project Work Phase I
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Background & Motivation

Instructions / Context

- The standard approach transfers raw pixels to a CPU/GPU over PCIe, introducing 5–50 ms of latency due to bus limits.
- Ballistic threats travel > 300 m/s. Every millisecond of delay translates to tracking errors.
- The total sensor-to-decision budget is under 500 microseconds.
- Cloud TPUs or edge GPUs have non-deterministic OS scheduling.

Target/Objectives

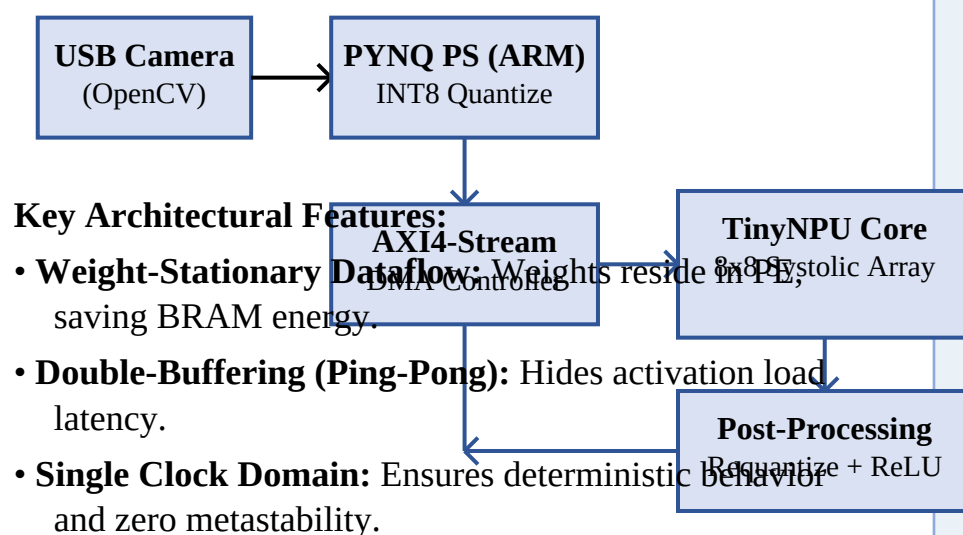
Primary Target

To achieve single-digit microsecond inference latency for Conv/FC layers on a resource-constrained Zynq-7020 platform.

Key Objectives

1. **Ultra-Low Latency:** Deterministic cycle-level timing using a custom hardware pipeline.
2. **Resource Efficiency:** Maximize the 220 DSP48E1 blocks and 140 BRAMs available on the Artix-7 fabric.

Basic Topology / Concept



Proposed Methodology

Proposed Methodology Details

The complete processing pipeline achieves high performance through parallel execution:

1. **Setup Phase:** The Python script running on the ARM PS preloads network weights into the FPGA's Weight BRAM via AXI4-Master DMA.
2. **Input Skewing:** To maintain systolic flow, incoming activations are skewed by 1 cycle per row.
3. **Computation:** The 8x8 MAC grid produces 64 partial sums (INT32) every clock cycle.

Expected Results & Performance

Hardware Resource Utilization (Zynq-7020)

- **DSP48E1 Slices:** 64 used (One per Processing Element)
- **BRAM:** 4–6 blocks (Ping-Pong Activation + Weight Buffers)
- **LUTs:** ~2,500 (Controller FSM, Requantizer, Address Decode)
- **Flip-Flops:** ~3,000 (Pipeline, Skew Registers, Output Staging)

Performance Projections

Conclusions

The TinyNPU project successfully demonstrates a near-sensor AI hardware accelerator capable of eliminating the traditional PCIe bottleneck in ballistic vision systems.

By utilizing an 8x8 weight-stationary systolic array and fully integrating it with a Python PYNQ frontend, the system provides:

- Deterministic, microsecond-level latency.
- High energy efficiency using INT8 quantization.
- A highly portable RTL design ready for ASIC migration.